Unit 2 Group Problem Session

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1 Instructions

Solve these problems together, in a group of four.
Each team member will write up the solution to one of the questions.
Submit your solutions to Gradescope (more instructions to follow in Canvas).
You will need to provide:

- The names of all team members
- Your solutions to the problems, indicating who wrote up each solution.
- **You may consult generative AI tools and other sources, but do not use them to write your solutions.**
- Cite any external sources or tools used and you used them, or indicate that you did not use any. **This part is required.**

The list of authors to our solutions for these questions and the subsequent usage of external tools or sources is listed as follows:

- Q1: Chris Amey - No external tools or sources
- Q2: Alex Hanus - No external tools or sources
- Q3: Nolan Knievel - AI was used for converting notes to LaTex
- Q4: Ronan Biggs - AI was used for converting notes to LaTex

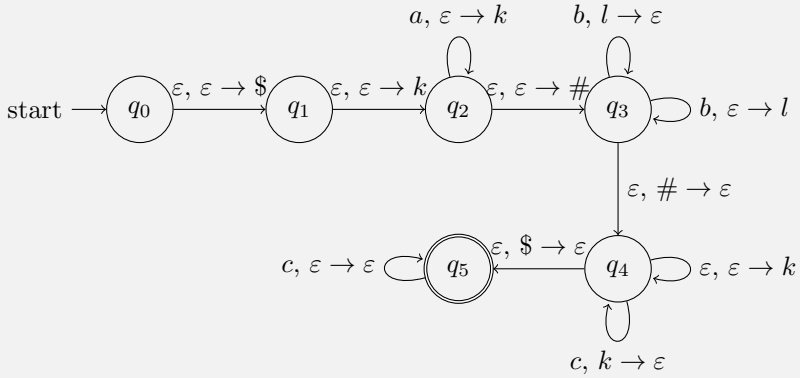
2 Questions

1. Let $\Sigma = \{a, b, c\}$. Draw a PDA that recognizes the following language:

$$\{a^k b^{2l} c^m \mid l \geq 0, m > k \geq 0\}$$

Author: Chris Amey

PDA:



AI Usage and Outside Sources: No AI agent or tool was used during the solution or writing of this problem, nor was the internet used in any way. The only thing referenced was the latex code for the Unit 1B group problem set for writing the the graph.

2. Write a CFG that generates the language

$$\{a^k b^{2l} c^m \mid l \geq k \geq 0, m \geq 0\}$$

Note this is NOT the language in the previous question.
The CFG does **not** have to be in Chomsky normal form (CNF).

Author: Alex Hanus

$$S \to AC$$
$$A \to aAbb|Abb|\varepsilon$$
$$C \to cC|\varepsilon$$

AI Usage and Outside Sources: No AI tools or outside sources

were used in the solving of this problem.

3. Consider the following CFG.

$$\begin{aligned} S &\to ZT \mid YR \mid ZZ \mid YY \mid 0 \mid 1 \\ X &\to ZT \mid YR \mid ZZ \mid YY \mid 0 \mid 1 \\ R &\to XY \\ T &\to XZ \\ Z &\to 0 \\ Y &\to 1 \end{aligned}$$

Consider the string 00100. Run the CYK algorithm to determine whether G generates this string. If it does, recover the parse tree that results from the algorithm. If not, say why not.

Author: Nolan Knievel

Answer: Yes, G generates 00100.

Length 1.

$$X_{1,1} = \{S, X, Z\} \qquad X_{2,2} = \{S, X, Z\} \qquad X_{3,3} = \{S, X, Y\}$$
$$X_{4,4} = \{S, X, Z\} \qquad X_{5,5} = \{S, X, Z\}$$

Length 2.

$$\begin{aligned} X_{1,2} &= (X_{1,1}, X_{2,2}) = \{S, X, Z\}\{S, X, Z\} \Rightarrow \{S, X, T\} \\ X_{2,3} &= (X_{2,2}, X_{3,3}) = \{S, X, Z\}\{S, X, Y\} \Rightarrow \{R\} \\ X_{3,4} &= (X_{3,3}, X_{4,4}) = \{S, X, Y\}\{S, X, Z\} \Rightarrow \{T\} \\ X_{4,5} &= (X_{4,4}, X_{5,5}) = \{S, X, Z\}\{S, X, Z\} \Rightarrow \{S, X, T\} \end{aligned}$$

Length 3.

$$\begin{aligned} X_{1,3} &= (X_{1,1}, X_{2,3}) \cup (X_{1,2}, X_{3,3}) \\ &= \{S, X, Z\}\{R\} \cup \{S, X, T\}\{S, X, Y\} \\ &\Rightarrow \emptyset \cup \{R\} = \{R\} \end{aligned}$$

$$\begin{aligned} X_{2,4} &= (X_{2,2}, X_{3,4}) \cup (X_{2,3}, X_{4,4}) \\ &= \{S, X, Z\}\{T\} \cup \{R\}\{S, X, Z\} \\ &\Rightarrow \{S, X\} \end{aligned}$$

$$\begin{aligned} X_{3,5} &= (X_{3,3}, X_{4,5}) \cup (X_{3,4}, X_{5,5}) \\ &= \{S, X, Y\}\{S, X, T\} \cup \{T\}\{S, X, Z\} \\ &\Rightarrow \emptyset \end{aligned}$$

Length 4.

$$\begin{aligned} X_{1,4} &= (X_{1,1}, X_{2,4}) \cup (X_{1,2}, X_{3,4}) \cup (X_{1,3}, X_{4,4}) \\ &= \{S, X, Z\}\{S, X\} \cup \{S, X, T\}\{T\} \cup \{R\}\{S, X, Z\} \Rightarrow \emptyset \\ X_{2,5} &= (X_{2,2}, X_{3,5}) \cup (X_{2,3}, X_{4,5}) \cup (X_{2,4}, X_{5,5}) \\ &= \{S, X, Z\}\emptyset \cup \{R\}\{S, X, T\} \cup \{S, X\}\{S, X, Z\} \Rightarrow \{T\} \end{aligned}$$

Length 5.

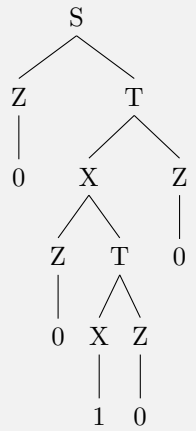
$$\begin{aligned} X_{1,5} &= (X_{1,1}, X_{2,5}) \cup (X_{1,2}, X_{3,5}) \cup (X_{1,3}, X_{4,5}) \cup (X_{1,4}, X_{5,5}) \\ &= \{S, X, Z\}\{T\} \cup \{S, X, T\}\emptyset \cup \{R\}\{S, X, T\} \cup \emptyset\{S, X, Z\} \\ \Rightarrow \{S, X\} \end{aligned}$$

CYK table:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$
length 5	$\{S, X\}$				
length 4	$\emptyset$	$\{T\}$			
length 3	$\{R\}$	$\{S, X\}$	$\emptyset$		
length 2	$\{S, X, T\}$	$\{R\}$	$\{T\}$	$\{S, X, T\}$	
length 1	$\{S, X, Z\}$	$\{S, X, Z\}$	$\{S, X, Y\}$	$\{S, X, Z\}$	$\{S, X, Z\}$
w	0	0	1	0	0

Since $S \in X_{1,5}$, G generates 00100.

Parse tree:



AI Usage and Outside Sources: I used AI to convert my handwritten solution to Latex.

4. Consider the language $L = \{r\#st\#ts \mid r, s, t \in \{0,1\}^*\}$. Prove that L is not context free using the pumping lemma for context-free languages.

Author: Ronan Biggs

Assume L is context-free. Then there exists a pumping length p. Let

$$w = r\#st\#ts, \qquad r = s = \varepsilon, \qquad t = 0^p 1^p.$$

Then $w \in L$ and $|w| \leq p$. $w = uvxyz$.

Case 1: # is in v or y. Let $w' = uv^2xy^2z$. Then the number of # symbols in w' is greater than two, so $w' \notin L$.

Case 2: uxy lies either between the # symbols or after the second #, and $|vy| > 0$. Let $w' = uv^2xy^2z$. Then

$$w' = \#t'\#t'', \qquad |t'| \neq |t''|.$$

Case 3: x contains the second #, and $|vy| > 0$. By (1), v contains only 1's and u contains only 0's. Let $w' = uv^2xy^2z$. Then $w' = \#t'\#t''$, where t' has more 1's than t'', or t'' has more 0's than t'.

By (2), these cases are exhaustive. By (1), $w' \in L$, but in all cases $w' \notin L$. This contradiction means that L is not context-free.

AI Usage and Outside Sources: I used AI to convert my written solution to LaTeX.